

Aadhaar App Intent Integration

Specification v2

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1. Introduction

This document provides a complete end-to-end guide for integrating with Aadhaar App for Verifiable Credential exchange.

The integration supports three invocation mechanisms:

1. App to App flow
2. Web to App flow
3. Cross Device QR flow

The JWT request structure remains common across all flows. Only the invocation method differs.

2. Before You Start

Before beginning integration, ensure the following prerequisites are completed.

2.1 Cryptographic Setup

You must generate an RSA key pair.

- 1) Private key → used to sign Intent Request JWT
- 2) Public key → shared with UIDAI during onboarding

Requirements:

- 1) Algorithm: RSA
- 2) Signing: RS256
- 3) Private key must remain in backend
- 4) Private key must never be embedded in frontend or mobile apps

2.2 OVSE Registration

To obtain OVSE Client ID, you must provide UIDAI:

- 1) RSA Public Key
- 2) Logo in SVG format (Less than 10 KB)
- 3) Callback URL
- 4) Domain Name
- 5) Entity Name to be displayed inside Aadhaar App

UIDAI will register your public key and issue OVSE Client ID. Integration cannot proceed without registration.

2.3 Backend Requirements

Your backend must support:

- 1) SD-JWT creation and parsing
- 2) RS256 signature verification
- 3) SHA256 hashing
- 4) Base64URL encoding
- 5) HTTPS callback endpoint
- 6) Secure private key storage

2.4 JWT Signature Validation (Recommended Pre-check)

Before sharing the public key and proceeding with integration, OVSEs are strongly recommended to validate the JWT signature format and structure.

Steps:

1. Generate a sample JWT using your private key
2. Verify the JWT on <https://jwt.io>
3. Ensure:
 - a. Algorithm is RS256
 - b. Signature is correctly verifiable using the public key
 - c. Payload structure matches UIDAI specification



This step helps identify implementation issues early and reduces integration failures.

(This is for development validation only and should not be used in production flows.)

3. Steps for intent generation

3.1 Create the Intent Request JWT

Every integration begins by constructing a signed JWT.

3.1.1 JWT Header

```
{  
  "alg": "RS256",  
  "typ": "JWT"  
}
```

Algorithm must be RS256.

3.1.2 JWT Payload Structure

```
{  
  "txn": "94f63917-708a-4e58-9001-fce56d839948",  
  "i": "credential",  
  "lang": "23",  
  "sc": "1110001110001110000001110001110001110001111",  
  "pop": 1,  
  "ch": "app",  
  "m": "1",  
}
```

```
{
  "ac": "1a2f",
  "sa": "00b1",
  "cb": "https://verifier.com/api/callback",
  "aud": "https://verifier.com",
  "iss": "https://uidai.gov.in",
  "exp": 1747094073,
  "iat": 1747093773,
  "ht": "RAM KUMAR",
  "aid": "in.gov.uidai.samvaad",
  "asig": "BASE64_SIGNATURE",
  "jti": "8a9d7c2b-b932-4131-9e36-43f22a019be5"
}
```

3.1.3 Payload Field Specification

Below is the authoritative field specification.

Field	Type	Format	Allowed Values	Mandatory	Applies To	What You Must Do
txn	String	UUID	Any valid UUID	Yes	All flows	Generate new UUID per request. Never reuse.
i	String	Literal string	"credential"	Yes	All flows	Must always be "credential".
lang	String	Numeric string	1-23	Yes	All flows	Pass numeric language code. Defaults to 23 if omitted.
sc	String	64-bit binary string	0 or 1 per bit	Yes	All flows	Construct full 64-bit bitmap. Set only

						required claims to 1.
pop	Integer	0 or 1	0 or 1	Yes	All flows	Set to 1. Face authentication required.
ch	String	Literal string	"app", "web", "qr"	Yes	All flows	Must match invocation method.
m	String	"0" or "1"	"0", "1"	Yes	All flows	Set to "1" for online authentication.
ac	String	Alphanumeric	OVSE Client ID	Yes	All flows	Use value issued by UIDAI if applicable.
sa	String	Alphanumeric	OVSE Registration number	Yes	All flows	Use code issued during onboarding.
cb	String	HTTPS URL	Valid URL	Yes	Web to App, Cross Device	Must be HTTPS and publicly accessible.
aud	String	URL (domain only)	Must match cb domain	Yes	Web to App, Cross Device	Must equal callback domain.
iss	String	URL	Fixed value	Yes	All flows	Use configured issuer value.
exp	Integer	Unix timestamp	Future epoch time	Yes	All flows	Set to short expiry, recommended 5 minutes.
iat	Integer	Unix timestamp	Current epoch time	Yes	All flows	Set to current timestamp.

ht	String	Plain text	Any string	Yes but can be a null value	All flows	Use as profile hint in multi-profile scenarios.
aid	String	Package ID	App ID	Yes Yes but can be a null value	App to App	Mandatory for App flow. Provide application ID.
asig	String	Base64 string	Encoded signature	Yes but can be a null value	App to App	Mandatory for App flow. Provide Base64 app signature.
jti	String	UUID	Any valid UUID	Yes	All flows	Generate unique JWT ID per request. Used for replay protection.

3.1.4 Complete Bitmap Mapping

You must always construct a 42 bit string

Bit	Claim	Description
1	credentialIssuingDate	Date on which the Verifiable Credential was issued
2	enrolmentDate	Date of Aadhaar enrolment
3	enrolmentNumber	Unique enrolment identifier assigned during registration
4	isNRI	Indicates whether resident is a Non Resident Indian (Boolean)
5	residentImage	Photograph of the Aadhaar holder
6	residentName	Full name of the resident in English
7	localResidentName	Full name of the resident in local language

8	ageAbove18	Indicates if resident is 18 years or older
9	ageAbove50	Indicates if resident is 50 years or older
10	ageAbove60	Indicates if resident is 60 years or older
11	ageAbove75	Indicates if resident is 75 years or older
12	dob	Date of birth of the resident
13	gender	Gender of the resident
14	careOf	Care of (C/O) field in English
15	localCareOf	Care of (C/O) field in local language
16	building	Building or house identifier (English)
17	localBuilding	Building or house identifier (local language)
18	locality	Locality or area name (English)
19	localLocality	Locality or area name (local language)
20	street	Street name (English)
21	localStreet	Street name (local language)
22	landmark	Nearby landmark (English)
23	localLandmark	Nearby landmark (local language)
24	vtc	Village, Town, or City (English)
25	localVtc	Village, Town, or City (local language)
26	subDistrict	Sub district / Tehsil (English)
27	localSubDistrict	Sub district / Tehsil (local language)
28	district	District name (English)
29	localDistrict	District name (local language)
30	state	State name (English)
31	localState	State name (local language)
32	poName	Post office name (English)
33	localPoName	Post office name (local language)
34	pincode	Postal PIN code
35	address	Full address in English
36	regionalAddress	Full address in regional language
37	mobile	Registered mobile number
38	maskedMobile	Partially masked mobile number
39	email	Registered email address
40	maskedEmail	Partially masked email address

41-42	Padding	Reserved for future use. Must be set to 0 unless specified
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3.1.5 Language Mapping Table

Code	Language
1	Assamese
2	Bengali
3	Bodo
4	Dogri
5	Gujarati
6	Hindi
7	Kannada
8	Kashmiri
9	Konkani
10	Maithili
11	Malayalam
12	Manipuri
13	Marathi
14	Nepali
15	Oriya
16	Punjabi
17	Sanskrit
18	Santhali
19	Sindhi
20	Tamil
21	Telugu
22	Urdu

3.2 Choose Invocation Method

3.2.1 App to App

Android:

```
val intent = Intent("in.gov.uidai.pehchaan.INTENT_REQUEST").apply {  
    putExtra("request", jwt)  
}  
startActivity(intent)
```

iOS:

```
UIApplication.shared.open(URL(string:  
"pehchan://in.gov.uidai.pehchan?req=\(jwt)")!)
```

3.2.2 Web to App

```
window.location.replace(  
"intent:#Intent;action=in.gov.uidai.pehchaan.WEB_INTENT_REQUEST;S.request=" + jwt + ";end"  
);
```

3.2.3 Cross Device QR

Steps:

1. Sign JWT
2. Encode ISO-8859-1
3. Append delimiter 255
4. Gzip compress
5. Convert to BigInt
6. Convert to Base10
7. Prefix: <https://maadhaar.com/getIntent?value=<BASE10>>
8. Generate QR

3.3 Receive SD-JWT

Callback format:

```
<Request>  
  <TxnID>TXN_ID</TxnID>  
  <Credential>SD-JWT</Credential>  
</Request>
```

Format:

a.b.c~d1~d2

3.4 Verify SD-JWT

3.4.1 Signature Verification

1. Split header.payload.signature
2. Retrieve UIDAI public key
3. Verify RS256 signature
4. Reject if invalid

3.4.2 Disclosure Verification

For each disclosure:

1. Hash disclosure string using SHA256
2. Base64URL encode
3. Match against _sd array
4. Reject if mismatch

Both validations are mandatory.

3.5 Send Acknowledgement

```
<Response>  
  <TxnID>TXN_ID</TxnID>  
  <ResponseCode>200</ResponseCode>  
  <ResponseMsg>Success</ResponseMsg>  
</Response>
```

For failure scenarios:

```
<Response>  
  <TxnID>TXN_ID</TxnID>  
  <ResponseCode><ERROR_CODE></ResponseCode>  
  <ResponseMsg><ERROR_MESSAGE></ResponseMsg>  
</Response>
```

3.6 Error Codes and Failure Scenarios

The following error codes standardize failure handling across Aadhaar App authentication flows. OVSE systems must handle these appropriately and provide meaningful user messaging.

Name	Case	Error Code	Error Message
User Reject	User explicitly taps Deny / Reject	301	User rejected the authentication request
Face Liveness Fail	Face liveness detection fails repeatedly, exceeding retry limit	302	Face liveness verification failed after 3 attempts
User Abort	User exits flow before completion (back press, app switch, close)	303	Authentication process aborted by user
Face Mismatch	Face captured successfully but does not match Aadhaar records	304	Face authentication failed due to mismatch with registered data
Biometric Lock	Biometric lock enabled in UIDAI system	305	Biometric authentication is locked for this user
FaceRD Server Down	FaceRD service unavailable	306	FaceRD service is currently unavailable. Please try again later
HTTP Error	Network/API failure (timeout, 5xx, invalid response)	307	Failed to complete authentication due to network or server error
Callback Retry	Auth success but callback delivery failed	308	Authentication completed but callback delivery failed. Retrying

3.7 Security Requirements

1. Private key must remain backend
2. Callback must use HTTPS
3. exp must be short lived
4. No credential logging
5. Validate txn and jti
6. Reject replayed transactions

3.8 Final Integration Checklist

1. RSA key pair generated
2. OVSE registered
3. JWT payload correctly constructed
4. Bitmaps correct
5. Language code corrects
6. JWT signed with RS256
7. Correct invocation method
8. SD-JWT verified fully
9. Acknowledgement sent